MIL-S-7867A(AER)
1 OCTOBER 1957
SUPERSEDES
MIL-S-7867(AER)
1 JANUARY 1952

MILITARY SPECIFICATION

SERVO ACTUATOR, D-9

This specification has been approved by
the Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall move aerodynamic control surfaces of target drones in response to radio command signals.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-T-152	Treatment, Moisture and Fungus Resistant, of Communications, Electronic and Associated Electrical Equipment
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-P-6064	Packaging of Lightweight Aircraft Accessories
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-M-7911	Marking, Identification of Aeronautical Equipment, Assemblies, and Parts
MIL-P-17555	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-T-18303	Test Procedures, Preproduction and Inspection for Aircraft Electronic Equipment; Format for
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment

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MIL-E-7894 Electric Power, Aircraft Characteristics of

Bureau of Aeronautics

SAR-300 Engineering Change Proposals, Electronic Material Changes and Electronic Material Bulletins; Preparation of

STANDARDS

MIL-STD-129 Marking for Shipment and Storage

2.2 Availability of Documents:

(1) Give both the title and number when requesting specifications, standards, drawings and publications. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania, Attention: Code ODPT.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specification are in conflict, the following precedence shall apply.

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the Bureau of Aeronautics.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern :

(1) Parts and materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the characteristics of standard MIL parts and materials are such that their use will not fulfill the requirements because of size, weight, performance or other reasons, material and parts shall be used which most nearly meet or exceed the requirements of the respective standard parts and material specifications.

(3) When a non-standard part is used, the meeting of general material requirements such as non-inflammability, fungus resistance, non-toxicity, etc., and, the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-E-5272 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass material, elec-

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trical and environmental tests. Any assembly built to be expended shall be approved first by the Bureau of Aeronautics.

3.1.1 Non-Standard Part and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-E-5400 except as follows:

(1) Mechanical parts shall be submitted as non-standard parts only when the material requirements of Specification MIL-E-5400 are not met.

(2) Non-standard parts of an electrical nature as defined in Specification MIL-E-5400 must be approved before the preproduction model of the equipment is submitted for tests. The request for approval must be forwarded to the Bureau of Aeronautics before the part is purchased or fabricated. In addition to or in lieu of the Non-Standard Part Data Sheet, the contractor shall submit an itemized list of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.2 Design and Construction:

3.2.1 General - This equipment shall fulfill all applicable design and construction requirements of Specification MIL-E-5400 unless otherwise specified herein.

3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the environmental and mechanical service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment.

3.2.3 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment:

Temperature	Room ambient 25°C +15°C or -5°C
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	27.5 ± 0.5 V DC

3.2.4 Total Weight - The total weight of the equipment including the control arm shall be a minimum consistent with good design and shall not exceed 2.5 pounds.

3.2.5 Operational Stability - The equipment shall operate with optimum performance for 5 hours, continuously or intermittently, without necessity for readjustment of any controls which are inaccessible during flight.

3.2.6 Operating Life

3.2.6.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 50 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.6.2 Total Operating Life - The equipment shall have a minimum total operating life of 100 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.7 Interconnection Cabling - This equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded, when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

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3.2.8 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with the requirements of Specification MIL-E-5400. However, cables and that portion of the connectors which is to be attached to the cables shall not be supplied as part of the equipment.

3.2.9 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in Specification MIL-E-7894 having limits as specified herein. The power required shall not exceed the specified amounts.

(1) DC Power 28 Volts System

Operating Voltage Limits	22.5 to 29 volts
Current	1.20 amps

3.2.9.1 Power Source Transients - The servo shall not be damaged when subjected to source transients and shall automatically resume normal operation at the expiration of the transient condition.

3.2.10 Grounded Input - The design of the servo shall be such that the negative supply to the motor shall be obtained from internal case ground.

3.2.11 Under Voltage Protection - The equipment shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

3.2.12 Moisture and Fungus Resistant Treatment - Equipments shall be fungus-proofed by selection of parts and materials as non-nutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that overall spraying of the equipment is not necessary. Overall spraying of the equipment in accordance with Specification MIL-T-152 shall be made in the event selection of parts and materials described above is not possible.

3.2.13 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.14 Interchangeability - The interchangeability requirements between like assemblies, subassemblies and replaceable parts as specified in Specification MIL-E-5400 shall be extended to require interchangeability of units of this equipment with the latest model of any previously produced equipment. Such interchangeability shall be measured against a model, manufacturing drawings, or other technical information as provided by the Bureau of Aeronautics for the purpose. In the event that requirements of Specification MIL-E-5400 conflict with the requirements for interchangeability, the interchangeability requirements shall govern.

3.2.15 Provisions for Maintenance and Field Testing - The equipment shall be designed to provide for ease of maintenance to the greatest practicable extent. Provisions for field testing shall be as specified in Specification MIL-E-5400.

3.2.16 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181 over the frequency range of 50 to 1000 megacycles.

3.3 Performance

3.3.1 Rated Load - The servo shall be capable of producing a minimum torque of 70 pound-inches with an output shaft speed of 8.5 rpm.

3.3.2 Non-Reversibility - The servo output shaft shall be non-reversible under torque loads equal to 30 pound-inches with no voltage applied to the motor.

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3.3.3 Stall Torque - The servo motor shall not stall under torque loads of up to twice rated load while operating on 26.5 volts dc.

3.3.4 Braking - The servo output shaft shall not coast more than 2° when the control signal is removed with the servo operating under 30 pound-inches aiding load.

3.3.5 Centering - Provision shall be made to center the output shaft upon application of an external command.

3.3.6 Speed - The servo shall have a no load speed of no greater than 10.5 rpm. The speed at rated load shall not be less than 7.5 rpm.

3.3.7 Endurance - The servo shall operate for a period of 50 hours when pulsed by commands at 15 cps operating on an "on-off" duty cycle of approximately "50-50%". The servo shall operate at the rate of approximately one complete oscillation every 5 seconds with the torque arm restrained by a spring whose tension is no less than 3 pound-inches per degree.

3.4 Workmanship - Workmanship, fabrication and manufacturing processes shall conform to the requirements of Specification MIL-E-5400.

3.5 Detail Requirements - The motor, gear train, and output shaft of the servo shall provide an output shaft speed of 8-1/2 rpm while delivering 70 inch-pounds load.

3.5.1 Form Factor and Mounting - The servo shall be contained in a waterproof case and shall incorporate four points of attachment located on the output shaft end.

3.5.1.1 Attachment Points - The four points of attachment shall be finished in a manner such as to assure a good electrical bond between the servo and the attaching surface.

3.5.2. Contents of Unit - The servo shall consist of the following major components:

(1) D.C. Motor

(2) Gear Train

(3) Output Shaft

(4) Control Arm

(5) Follow-up Potentiometer

(6) Relays

(7) Arc Suppression Network

(8) Limit Switch

(9) Centering Switch

(10) Receptacle.

3.5.2.1 Motor - A direct-current electric motor, designed to operate on 22.5 to 30 volts, shall be provided and shall be equipped with oil or graphite impregnated bearings, or grease packed ball bearings. The motor shall be suitably geared to the output shaft to give the performance specified.

3.5.2.2 Gear Train -

3.5.2.2.1 Mechanical Strength - The gear train and bearings shall be capable of withstanding shock loads on the control arm equivalent to a weight of 5.5 pounds dropping 5 inches on

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the 1.45 inch control arm station with 100 pound-inches already on the arm.

3.5.2.2.2 Backlash - Gear train backlash shall be held to less than 1° total for the servo as measured by traversing the control arm.

3.5.2.3 Output Shaft - The output shaft shall be provided with standard 36 S. A. E. serrations.

3.5.2.4 Control Arm - A control arm shall be provided with standard 36 S. A. E. serrations similar to those on the output shaft.

3.5.2.5 Follow-up Potentiometer - A precision wirewound potentiometer of 1-watt rating shall be furnished as an integral part of the servo. This unit shall have a total resistance of 1200 ohms, ± 4%, with a linearity of 1/2% of full scale, and a resolution of 1/2°. The windings shall cover an arc of 50 + 3 - 0°, and have a 10° override on each end of the windings without causing an open circuit. The wiper shall be connected to the output shaft so that it rotates at the same speed as the control arm.

3.5.2.6 Relays - Two relays capable of operating through a range of 18 to 30 volts DC, shall be provided as a part of the servo to reverse the motor. The relay actuating current shall be less than 0.25 ampere of direct current.

3.5.2.7 Arc Suppression - The servo shall contain an arc suppression network to protect the relay contacts from burning, pitting or welding when operating under any of the conditions specified herein.

3.5.2.8 Limit Switch - A combination limit switch shall be provided so that the control signal shall be automatically released when the output shaft has rotated to a preset clockwise or counterclockwise position.

3.5.2.8.1 Control Arm Travel - The limit switch shall be adjusted to limit the travel of the control arm to 22.5° ± 2.5° either side of center when the control voltage is impressed on the motor.

3.5.2.9 Centering - The mechanical center of the control arm travel, as determined by the centering switch, shall be adjusted so that with the servo centered, the potentiometer wiper shall be within ± 1° of the electrical center of the potentiometer.

3.5.2.9.1 Positioning of Control Arm - After the servo has completely responded to an external centering command, the output shaft shall be positioned so that the control arm may be installed with its center line parallel to the line connecting two diametrically opposite mounting holes within ± 2°.

3.5.2.10 Receptacle - The servo shall be provided with an AN3102C-16S-1P receptacle. The receptacle shall be located in the same end of the servo as the output shaft.

3.5.2.10.1 Interconnecting Cable - The interconnecting cable and that portion of the connectors which are to be attached to the cable shall not be supplied as part of the servo.

3.5.2.10.2 Wiring to Receptacle - The wiring to the receptacle shall be as follows:

(1) Clockwise rotation is defined as clockwise movement of the output shaft when viewed from the shaft and connector end of the servo.

(2) Case to be grounded to provide negative power connection.

Pin A - Counterclockwise Rotation Command Signal (Ground Signal).

Pin B - Clockwise Rotation Command Signal (Ground Signal).

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Pin C - Power (Nominal 26. 5 vdc.)

Pin D - Centering Command (Ground Signal)

Pin E - Potentiometer (end towards which wiper moves when shaft rotates CW).

Pin F - Potentiometer (end towards which wiper moves when shaft rotates CCW).

Pin G - Potentiometer wiper.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine conformance with all the applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting these tests shall be submitted to the Bureau of Aeronautics for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory satisfactory to the Bureau of Aeronautics.

4.2.2 Service Approval Tests - The service approval model shall consist of four equipments representative of the production equipment to be supplied under the contract. This may be the same equipment that was subjected to the contractor's demonstration tests. These equipments shall be delivered to the destination specified by the Bureau of Aeronautics. When the tests of 4.2.1 are supervised, the Bureau of Aeronautics may, when conducting service approval tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting service approval tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. The design and test data shall include the data obtained by the contractor in testing to Specification MIL-E-5272. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-E-5272 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

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4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Bureau of Aeronautics shall engage the service of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements of Specification MIL-E-5400 have been met.

4.3.1.2 *Operational Test - Each servo shall* be operated for a period of 30 minutes and at the end of this time shall operate within the performance limits of this specification. During this 30 minute period, the servo shall be operated at 22.5 V DC for 5 minutes, 30 V DC for 5 minutes, and 26.5 V DC for 20 minutes. The servo shall be operated as follows:

(1) The servo shall be pulsed at 15 cps ± 10%. The "on-off" duty cycle shall be approximately "50-50%".

(2) The servo shall be operated at a rate of approximately 1 complete oscillation every 5 seconds.

(3) Movement of the torque arm shall be resisted by a spring whose tension is no less than 3 pound-inches per degree.

Immediately after this test, the servo shall be subjected to the following:

4.3.1.2.1 Current Drain - The servo shall be operated in both directions under a torque of 70 pound-inches applied to the control arm. The running current shall not exceed 1.2 amperes.

4.3.1.2.2 Speed - The servo shall have no load speed of no greater than 10.5 rpm. The speed at rated load shall not be less than 7.5 rpm.

4.3.1.2.3 Non-Reversibility - The servo shall be securely fastened to a bench and a torque of 30 pound-inches applied to the control arm. With no voltage applied to the servo motor, the servo shall not turn.

4.3.1.2.4 Braking - *With the servo operating under* a torque of 30 pound-inches aiding, applied on the control arm, the control signal shall be released at any random point on the full range of travel. Angle of coasting shall be measured and shall not exceed 2° at any time.

4.3.1.2.5 Immersion - Each servo, while at room temperature, shall be submerged in 2 feet of water at a temperature of 49°C, for a period of not less than 4 minutes. There shall be

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no sign of leakage as evidenced by air bubbles. A cap or mating plug may be used with the connector during this test.

4.3.1.2.6 Centering - After the servo has responded to an external centering command, the wiper shall be within ± 1° of the electrical center of the potentiometer.

4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests.

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2
101 to 200	1
Over 200	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

4.3.2.1.1 Altitude - Procedure I of MIL-E-5272 except that the altitude shall be 30,000 feet for a period of 8 hours.

4.3.2.1.2 Endurance - The servo shall be subjected to pulsed commands for a period of 50 hours. The first 8 hours shall be run in compliance with the altitude requirements of 4.3.2.1.1. At the end of 50 hours, the servo shall operate within the performance limits of this specification. The servo shall be operated as follows:

(1) The servo shall be pulsed at 15 cps ± 1%. The "on-off" duty cycle shall be approximately "50-50%".

(2) The servo shall be operated at a rate of approximately one complete oscillation every 5 seconds.

(3) Movement of the torque arm shall be resisted by a spring whose tension is no less than 3 pound-inches per degree.

4.3.2.1.3 Radio Noise - Radio noise tests shall be conducted in accordance with Specification MIL-I-6181, except that measurements shall be made only between the range of 50-1000 megacycles.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling test.

4.3.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the Bureau of Aeronautics)

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Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the Bureau of Aeronautics. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted.

(1) High and Low Temperature - Procedure I of Specification MIL-E-5272.

(2) Humidity - Procedure I of Specification MIL-E-5272.

(3) Vibration Test (90 minutes cycling - in each of 3 planes)

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the Bureau of Aeronautics for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Reconditioning of Test Samples - An equipment which has been subjected to the tests shall be reconditioned as follows:

(1) On completion of the tests, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the Bureau of Aeronautics.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.5 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the Bureau of Aeronautics for review and approval. The right is reserved by the Bureau of Aeronautics or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

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4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring agency.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required, the contractor shall also prepare an Electronic Material Change (EMC) and an Electronic Material Bulletin (EMB) to cover the approved change. ECP's, EMC's and EMB's shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555 and MIL-STD-129.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 26.5 ± 1.0 V DC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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